

A review on thermal analysis of industrial boilers.

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Abstract. Industrial boilers are crucial components in various industrial processes, providing the necessary steam or heat for operations. The thermal performance of these boilers significantly impacts their efficiency, operational costs, and environmental footprint. The thermal analysis of industrial boilers involves examining how these systems handle and manage heat. Understanding their thermal behaviour is essential for efficiency, safety, and reliability. This review paper explores the key aspects of thermal analysis in industrial boilers, summarizing recent research findings and methodologies used to improve boiler performance.

Keywords. Thermal Analysis, Industrial Boiler, Comparisons

1. Introduction

The global industrial boilers market is expected to reach \$14.6 billion by 2028 from \$11.5 billion in 2023 at a CAGR of 4.8%[1]. With increasing demands of boilers from transition to manufacturing industries, it is important to consider the efficiency and the performance of the boiler to get the maximum output and long-term use without any damage or interruption[2]. Even though industrial systems are very diverse almost everyone had a steam-boiler that is used in one of its process.[3] Industrial boilers are the workhorses of various industries, providing essential heat and steam for manufacturing processes. Understanding the different types of boilers, their key components, and factors to consider during selection is vital for optimising performance and efficiency. From 1875 A.D, use of steam boilers has been found to be used on power stations [4]. Industrial boilers emerged in the late 18th century with the rise of steam engines during the Industrial Revolution. Initially simple, they evolved significantly by the 19th and 20th centuries, adopting multitube and water-tube designs for higher capacities. Innovations in the mid-20th century, such as packaged boilers and automated controls, improved safety and efficiency. Today, industrial boilers focus on energy efficiency, emissions reduction, and the use of alternative fuels, reflecting ongoing advancements in sustainability and technology [5].

1.1 Types

A type is Fire-tube boilers which are cost-effective, fuel-efficient, and easy to operate, with capacities up to 25 tons/hr. and pressures of 17.5 kg/cm². [4] Water tube is just opposite of fire tube boiler where hot gasses are passed through tubes which are surrounded by water. In water-tube boilers, water and steam flow inside the tubes while hot gases flow over the outside surface. Modern high-capacity boilers use this design due to their higher pressures and capacities compared to fire-tube boilers [4]. The biomass fuelled, stoker-fired, natural circulation drum-type steam generator has a full load capacity of 54.4 kg/s superheated steam at 106.9 bar and 786 K. A steam drum receives feed water from the economizer outlet that mixes with the saturated water in the drum. This sub-cooled mixture flows out of the drum through two parallel down comers that branch to feed the inlet headers of steam generating banks in the convective pass. The down comers also supply the lower water wall inlet headers, which feed the front,

side, and rear water wall tubes[5]. Waste Heat Recovery Boilers (WHRB) improves energy efficiency but also reduces operational costs and emissions. These boilers can be of either fire tube or water tube. The specific objective in this review is to have a comparative understanding in the current trends for thermal analysis of these boilers and a review on correlation between material use, heat transfer rates, thermal efficiency, combustion analysis, boiler dynamics, heat losses & temperature distribution across the boiler and studies.

2. Methodologies

The methodologies for thermal analysis of industrial boilers are diverse, including experimental methods, computational simulations, and theoretical models. This section will detail the various methods used in the studies reviewed till 2024.

2.1 Literature review

The current state of thermal analysis techniques applied to industrial boilers, encompassing exergy analysis, thermal efficiency optimization, computational fluid dynamics (CFD) simulations, and circulation analysis. Exergy analysis is a robust method for identifying inefficiencies within boiler systems. An exergy analysis provides a detailed understanding of where and how energy losses occur within a boiler system. FEA is used to calculate the temperature field in the boiler walls. This method involves solving the heat conduction equations for the boiler geometry using numerical techniques. A study implemented exergy analysis on industrial boilers to pinpoint areas where energy conservation measures could be most effective. They concluded that considering both energy and exergy aspects provides a holistic view of boiler performance, which is crucial for identifying improvement opportunities Energy Policy [9]. Thermal Efficiency and Optimization Improving the thermal efficiency of industrial boilers is a primary objective for many industries aiming to reduce fuel consumption and emissions. An extensive analysis to identify the main factors influencing boiler energy consumption and proposed design modifications to enhance thermal efficiency emphasized the importance of optimizing boiler design and operational parameters to achieve significant energy savings [10]. The integration of CFD with field data provides a sophisticated approach to analysing thermal performance in industrial boilers. CFD simulations, combined with field data, could accurately model and analyse the thermal behaviour of oil/gas boilers. The approach allows for precise adjustments and optimizations, leading to improved boiler performance and reduced emissions [11]. A research focused on energy-exergy analysis of coal-fired industrial boilers, using thermal test data to develop an exergy analysis model tailored for these boilers, highlighted the necessity of specific models for different boiler types to accurately assess and enhance their thermal performance. They found that energy-exergy analysis could identify key areas for efficiency improvement, leading to more effective energy use and reduced environmental impact [12]. Proper boiler circulation is essential for efficient heat transfer and boiler operation. A comprehensive approach to boiler circulation analysis, addressing all aspects of boiler circulation circuits was conducted with methodology ensuring that all circulation aspects are thoroughly analysed, leading to improved boiler reliability and performance [13]. This systematic approach is crucial for maintaining optimal heat transfer and preventing common issues such as overheating or inefficient fuel use Heat Transfer Engineering.

2.2 Factors affecting boiler efficiency

Several factors influence the efficiency of industrial boilers, including fuel type, combustion conditions, heat transfer surfaces, and operational practices. Understanding these factors is crucial for optimizing boiler performance and reducing energy consumption. Key factors affecting boiler efficiency include, type and quality of fuel used in a boiler significantly impact its efficiency. Fuels with higher heating values and lower moisture content typically result in higher efficiency. Proper air-fuel ratio, adequate mixing, and complete combustion are essential for maximizing boiler efficiency. Incomplete combustion leads to energy losses and increased emissions. The design and cleanliness of heat transfer surfaces play

a critical role in heat absorption and transfer. Fouling and scaling on heat transfer surfaces reduce efficiency by insulating the heat transfer process. Efficient boiler operation requires proper maintenance, regular inspections, and optimization of operating parameters such as pressure, temperature, and load conditions. Each type of boiler has its advantages and limitations, making them suitable for different applications. Fire-tube boilers are favoured for their simplicity and lower costs in smaller installations, while water-tube boilers are preferred in large-scale industrial settings due to their high efficiency and capacity. Electric boilers, although more expensive to operate, provide a clean and efficient solution for specific applications where emissions and space constraints are a concern.

2.3 Thermal analysis techniques

There are various experimental methods with which thermal analysis can be performed. Heat Balance Method involves calculating the heat transfer coefficient by measuring the total heat power absorbed by the boiler walls and dividing it by the surface area and temperature difference between the bulk of the bed and the tube surface. Local Heat Flow Meters are inserted through holes in the boiler walls to measure local heat flux directly. The heat transfer coefficient is then calculated based on the measured heat flow and the temperature difference between the bulk of the bed and the meter surface. Thermocouple Measurements are used to measure the temperatures at various points on the boiler walls. These measurements are used to infer the local heat transfer coefficients by comparing the temperatures of the fin and tube surfaces. Water Temperature Method involves measuring the temperature increase of the cooling water as it flows through the boiler tubes. The heat transfer coefficient is calculated based on the heat absorbed by the water and the temperature difference between the bulk of the bed and the tube surface [8].

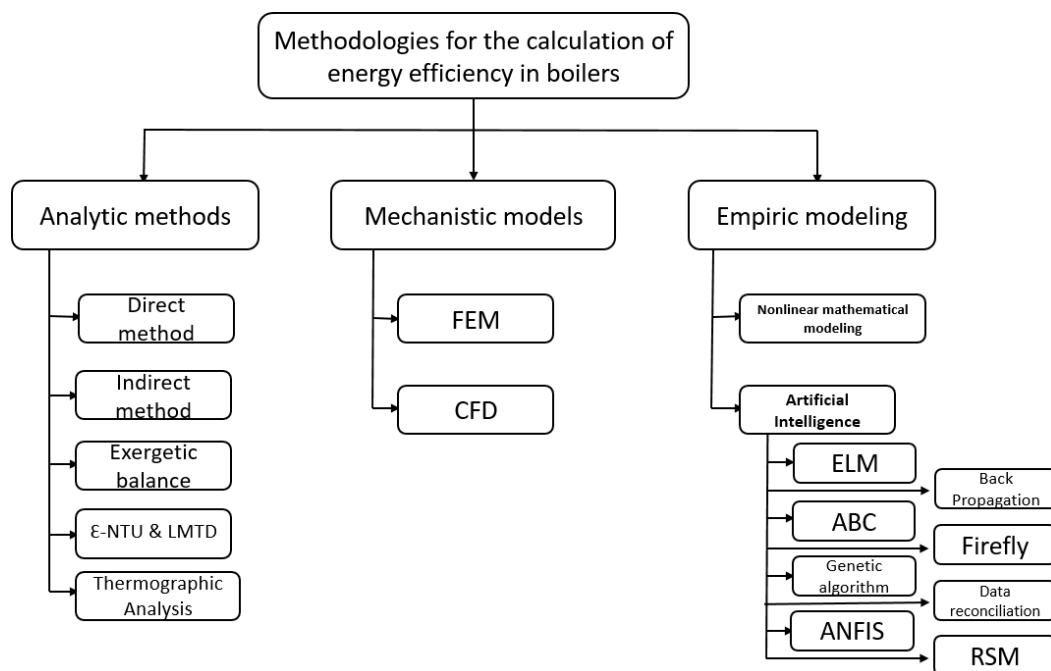


Figure 1. Methodologies for the calculation of energy efficiency in boilers.[9]

2.3.1 Analytical methods

This category includes methods that determine the efficiency of a part of the boiler or the entire one,

based on models constructed from energy balances, mass balances, exergy balances, and heat transport equations. The two most important methods in this category are both those that allow the calculation of the so-called direct efficiency and indirect efficiency, which are standardized methods in the Performance Test Code 4 (PTC 4) of The American Society of Mechanical Engineers ([ASME], 2013).

2.3.1.1 Direct method

The direct method calculates the efficiency by comparing the energy present in the steam output stream and the useful energy present in the fuel used, as shown in Equation 1 (Lang, 2009).

$$\eta_c = \frac{Q_w}{Q_c} * 100 \quad [\%] \quad (1)$$

In Equation 1, η is the direct boiler efficiency, Q is the energy in the steam stream, and Q is the energy available in the fuel. To determine the useful energy present in the fuel, it is necessary to know its heating value [6] The gross calorific value (GCV) is used, which includes the heat lost by vaporization of the water in the products.

Determining efficiency from the direct method requires accurate and direct measurements of multiple variables. The main variables required are the inlet water flow, the outlet water flow, the secondary outlet flows (blowdowns and auxiliary streams), the pressures and temperatures of the different streams, the fuel flow, and the gross calorific value of the fuel. The calculation of the energy in the steam stream is carried out through Equation 2.

$$Q_w = \sum Mr_{S_{z2}} (H_{S_{z2}} - H_{E_{z1}}) \quad (2)$$

In Equation 2, $H_{E_{z1}}$ is the specific enthalpy of the fluid entering the system, $H_{S_{z2}}$ is the specific enthalpy of the fluid leaving the system and $Mr_{S_{z2}}$ is the mass flow leaving the system. The energy available in the fuel is calculated according to Equation 3.

$$Q_c = MrF * GCV \quad (3)$$

In Equation 3, GCV is the specific gross calorific value, and MrF is the fuel mass flow rate (ASME, 2013) GCV is calculated according to Equation 4.

$$GCV = \sum_{i=1}^n \% (X_i)_{mole} \frac{M_i}{MW_F} GCV_i \quad (4)$$

M_i , $\% (X_i)_{mole}$ and GCV_i are molar mass, mole fraction, and GCV of the individual components, and MW_F is the molar mass of the fuel.

2.3.1.2 Indirect method

The indirect method calculates the energy losses in the boiler and subtracts them from efficiency of 100 %. Equations 5 and 6 present the calculation of the indirect efficiency.

$$\eta_c = 100 - \sum q_n \quad (5)$$

$$\sum q_n = q_1 + q_2 + q_3 + q_4 + q_5 + q_6 + \dots \quad (6)$$

Equations 5 and 6, η is the indirect boiler efficiency, and q corresponds to the different loss terms to be considered in the calculation. For example, the heat loss in the flue gas is calculated according to Equation 7.

$$q_1 = \frac{m_{FG} C_{p_{FG}} (T_{FG} - T_a)}{GCV} * 100 \quad [\%] \quad (7)$$

Where m_{FG} is the gas mass flow at the outlet, C_{pFG} is the specific heat of the fuel, and T_a is the ambient temperature.

The number of loss terms can vary according to the level of detail of the modeling. Bujak et. al performed modeling with nine loss terms, including some particularly associated with the use of coal or coal crushing gas fuel. Rehan et. al. used modeling with losses from flue gas, moisture in air and fuel, partial combustion of coal in CO, and radiation and convection. ASME PTC 4.1 uses the following losses: dry flue gas leaving, moisture in the flue gas, moisture in the combustion air, radiation at the boiler surface, and blowdown (ASME, 2013). Heuristic considerations are sometimes used to assign values to some of the loss terms. For example, A study [7] assigned constants 0.015, 0.04, and 0.005 to the last three-loss terms listed previously, based on the 2008 PTC 4.1. The largest energy losses are considered to occur in combustion, heat exchanger, and flue gas [8].

For the indirect efficiency calculation, measurements are required to calculate the loss terms included in the modeling, which implies the need of measuring flue gas temperature, GCV, excess air, thermal properties of flue gas components, temperature, pressure, and ambient humidity.

The flue gas temperature is a consequence of energy that ends up heating a non-used stream, although economizers are sometimes used to recover part of this energy. Calculation of the energy lost in the flue gas requires a calculation of the calorific value (C_p) of the stream, which in turn depends on the C_p of the components of the stream. These C_p can be calculated with Equations 8, 9, 10, 11, and 12, where the C_p is given in kJ/kmol*K, for temperatures given in K.

$$C_{p, CO_2} = -3.7357 - 4.1034 \left(\frac{T}{100}\right) + 30.529 \left(\frac{T}{100}\right)^{0.5} + 0.024 \left(\frac{T}{100}\right)^2 \quad (8)$$

$$C_{p, H_2O} = 143.05 - 183.54 \left(\frac{T}{100}\right)^{0.25} + 82.751 \left(\frac{T}{100}\right)^{0.5} - 3.6989 \left(\frac{T}{100}\right) \quad (9)$$

$$C_{p, N_2} = 39.060 - 512.79 \left(\frac{T}{100}\right)^{-1.5} + 1072.7 \left(\frac{T}{100}\right)^{-2} - 820.4 \left(\frac{T}{100}\right)^{-3} \quad (10)$$

$$C_{p, O_2} = 37.432 + 0.02 \left(\frac{T}{100}\right)^{1.5} - 178.57 \left(\frac{T}{100}\right)^{-1.5} + 236.88 \left(\frac{T}{100}\right)^{-2} \quad (11)$$

$$C_{p, CO} = 69.145 - 0.704 \left(\frac{T}{100}\right)^{0.75} - 200.77 \left(\frac{T}{100}\right)^{-0.5} + 176.76 \left(\frac{T}{100}\right)^{-0.75} \quad (12)$$

The specific heat of the gas corresponds to Equation 13.

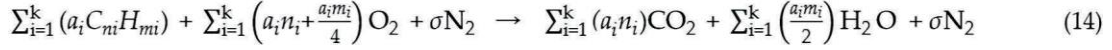
$$C_{p_{FG}} = \sum_{i=1}^k \%X_{i,mol} C_{pi} \quad (13)$$

The $C_{p_{FG}}$ is converted from molar to mass basis using the molecular weight of the gas.

Excess air is the additional amount of air to that stoichiometrically required, which is added to ensure complete combustion. The minimum amount of air is calculated according to the stoichiometry of the combustion reactions, considering that the air contains 21 % oxygen; nevertheless, in practice, complete combustion is not achieved by supplying the minimum air, because the mixture between air and fuel is not perfectly homogeneous, to the low residence time in the chamber, and kinetic issues of the reactions. It may cause incomplete combustion, which results in the generation of carbon monoxide (CO), and negatively influences efficiency. However, excess air also affects energy efficiency, as the incoming air is heated, consuming energy.

For the calculation of the Air-Fuel Ratio (AFR), a generic combustion reaction is shown in

Equation (14).



Where a_i , n_i and m_i are the stoichiometric coefficient and amount of moles of carbon and hydrogen for the i -th species in the fuel, respectively. The theoretically required amount of nitrogen (air) is given by σ in Equation 15:

$$\sigma = \frac{79}{21} \sum_{i=1}^k \left(a_i n_i + \frac{a_i m_i}{4} \right) + \sigma_F \quad (15)$$

To calculate the excess air required, the term βO_2 is added in products for the combustion reaction, so that the moles of nitrogen, σ , change to Equation 16.

$$\sigma = \frac{79}{21} \left(\sum_{i=1}^k \left(a_i n_i + \frac{a_i m_i}{4} \right) + \beta \right) \sigma_F \quad (16)$$

2.3.1.3 Exergetic balance

Exergy is the maximum amount of work available from a flow and is calculated by bringing the flow to a thermodynamic equilibrium state with reference (ambient) conditions. An exergy balance simultaneously assesses the quantity and quality of energy associated with the [9]. It can also be interpreted as a measure of the energy irreversibilities associated with this process since unavoidable energy losses are not energetically equivalent [10]

A performed a parametric analysis of destruction and efficiency concerning reference conditions $T = 25^\circ C$ and $P = 1$ bar, they found that for an increase in temperature, the destruction increases and the efficiency decreases. The calculation associated with exergy can be divided into subsystems: exergy destruction in the boiler, convection losses, destruction in the heater, loss in the emitted gas, loss due to CO formation, and loss due to unburned fuel.

Like energy efficiency, the greatest exergy loss occurs at the burner, followed by the heat exchanger. The blowdown is not considered a loss, but a product of exergy as it depends on the quality of the water and not on the efficiency of the boiler. Inside the air mixer, there is no energy loss, but there is an exergy loss associated with mixing and heat transfer. Kinetic and potential energy are not considered in these balances. Losses due to radiation and incomplete combustion are negligible for a properly functioning system but should be regarded if the burner or insulation is considered to warrant it. Briefly, the balance could be reduced to product exergy, fuel exergy, losses, and destruction.

To obtain the exergy losses, mass, energy, and exergy balances must be established, leading to Equation 17.

$$\dot{E}_F = \dot{E}_p + (\dot{E}_{L1} + \dot{E}_{L2} + \dot{E}_{L3} + \dot{E}_{L4}) + (\dot{E}_{D1} + \dot{E}_{D2}) \quad (17)$$

$\dot{E}_F$ is fuel exergy, $\dot{E}_p$ is exergy in products, $\dot{E}_{L,i}$ terms are exergy losses and $\dot{E}_{D,j}$ terms represent exergy destruction. $\dot{E}_F$ contains 3 components, as shown in Equation 18.

$$\dot{E}_F = \dot{E}_f + \dot{E}_{AS} + \dot{E}_{a,11} \quad (18)$$

$\dot{E}_F$ is the chemical exergy of the fuel consumed, $\dot{E}_{AS}$ is the exergy of the atomized stream, and $\dot{E}_{a,11}$ is the physical exergy of the air.

$\dot{E}_p$ can be calculated according to Equation 19, where $\dot{m}_i$ are the mass fluxes of the different component ε_i are the exergies of the components. Among the exergy losses, $\dot{E}_{L1}$ is that associated with the stack gas, according to Equation 20.

$$\dot{E}_P = \sum \dot{m}_i * \varepsilon_i \quad (20)$$

M_G is the equivalent molecular mass of the gas, product of the molar mass of the components multiplied by their mole fraction, and is the mass of gas measured at the stack exit. $\dot{E}_{L2}$ is the exergy associated with the unburned fuel, which for the case of gaseous fuels is assumed to be zero. $\dot{E}_{L3}$ is the exergy associated with the incomplete combustion emission and CO formation.

For the case of $\dot{E}_{L4}$, associated with exergy dissipation through the boiler surface, it is calculated with Equation 21 (ASME, 2013).

$$\dot{E}_{L4} = Q_{radiation} * \left(1 - \frac{T_0}{T_s}\right) \quad (21)$$

Where T_0 is the ambient temperature and T_s is the boiler surface temperature.

As for the exergy destruction terms, two aspects are considered: one is associated with the air heater, which is not considered for all types of boilers. The other corresponds to the exergy destroyed in the boiler, according to Equation 22 [9].

$$\dot{E}_{DI} = \dot{E}_F - \dot{E}_p - \left(\frac{\dot{m}_G}{M_G} * \tilde{\varepsilon}_{ch} + \dot{m}_G * \tilde{\varepsilon}_{ph} \right) - \dot{E}_{L4} \quad (22)$$

A direct exergy efficiency can be calculated using the exergy losses according to Equation 23.

$$\eta = \frac{E_p}{E_F} \quad (23)$$

Alternatively, the indirect exergy efficiency can be calculated with Equation 24.

$$\eta = 1 - \frac{\text{destroyed exergy}}{\text{incoming exergy}} = 1 - \frac{\sum(\dot{E}_L \dot{E}_D)}{\dot{E}_F} \quad (24)$$

2.3.1.4 LMTD and $\mathcal{E}$ -NTU

The Logarithmic Mean Temperature Difference (LMTD) is used to estimate the temperature associated with heat transfer in heat exchangers. Thus, this approach can be used to calculate the efficiency of the heat exchanger subsystem in a boiler. The LMTD is a logarithmic approximation of the temperature difference between the heat exchanger inlets and outlets; its value is directly proportional to the heat transferred and can be calculated from Equation 25.

$$LMTD = \frac{\Delta T_A - \Delta T_B}{\ln\left(\frac{\Delta T_A}{\Delta T_B}\right)} = \frac{\Delta T_A - \Delta T_B}{\ln \Delta T_A - \ln \Delta T_B} \quad (25)$$

Where ΔT_A is the temperature difference between the two streams at point A and ΔT_B is the temperature difference at point B. The number of units transferred (NTU) method is an alternative for estimating the heat transfer rate in heat exchangers when not all the data needed to do so through LMTD are known. The NTU method can be applied regardless of the flow distribution, either parallel or counterflow (Trojan, 2019), according to Equations 26, 27, and 28.

$$\varepsilon = \frac{1 - \exp\left(-NTU \cdot \left(1 - \frac{\dot{C}_{min}}{\dot{C}_{max}}\right)\right)}{1 - \frac{\dot{C}_{min}}{\dot{C}_{max}} \exp\left(-NTU \cdot \left(1 - \frac{\dot{C}_{min}}{\dot{C}_{max}}\right)\right)} \quad (26)$$

$$NTU = \frac{k \cdot A}{\dot{C}_{min}} \quad (27)$$

$$\dot{C}_{min} = \min \left[(C_p \dot{m})_{gas}, (C_p \dot{m})_{steam} \right] \quad (28)$$

Where $\dot{C}_{min}$ is the minor between steam and flue gas flow, k is the transfer constant and A is the heat exchange surface area [11]

2.1.3.5 Thermographic analysis

The thermographic analysis is e to establish radiation losses, using a thermal camera to record boiler surface temperatures. This allows estimating heat transfer losses in the walls of the boiler, radiation losses in different areas of the boiler, and detecting insulation problems. This analysis is important insofar as radiation losses are energy that is directly dissipated to the environment, diminishing the efficiency of the boiler. The way to perform the analysis is based on the sum of the surface areas where the temperature is above the expected. The analysis also can be carried out by determining what percentage of the surface area has such unacceptable or worrying temperatures. These situations can be caused by poor burner adjustment or poor insulation. The interpretation of the histogram is done through areas; the total area with a certain temperature is directly proportional to the histogram height for the same temperature [12] Figure 2 illustrates the process of loss analysis by thermography.

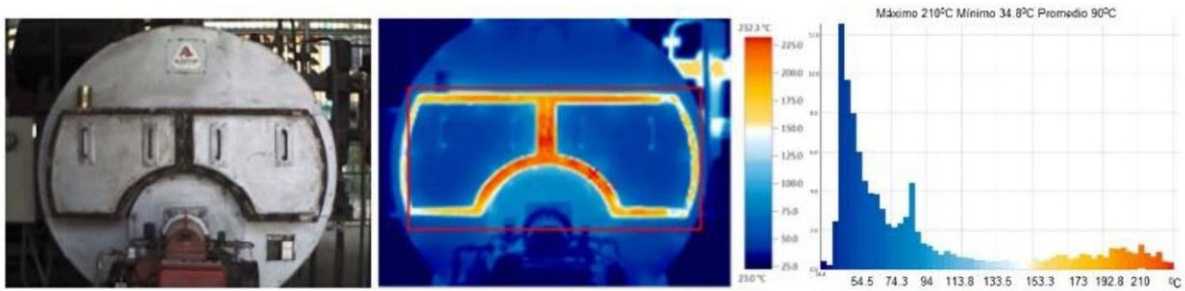


Figure 2. Boiler front surface, thermographic camera shot, and histogram with the corresponding distribution [13]

2.3.2 Mechanistic Models

Mechanistic models divide a complex system into subsystems or parts. To understand the behavior of the components of a system, mechanistic models assume that a system can be understood by analyzing how the different parts perform together and separately. Typically, mechanistic models are associated with a physical, tangible system whose components are solid and visible.

2.3.2.1 FEM

The Finite Element Method, FEM, is a numerical method employed in the solution of partial differential equations associated with engineering and physics problems. It is used in the design of industrial applications and the simulation of complex physical systems. The development of a FEM algorithm to solve a problem requires a four-stage development: (1) formulation of the problem in a variational form, (2) division of the domain of independent variables into finite elements, (3) projection of the original variational problem onto the vector space and (4) numerical computation of the solution of the system of equations. These steps allow representing a differential calculus problem with a linear algebra problem. The problem is posed on a vector space of non-finite dimension; nevertheless, it can be solved approximately by finding a projection onto a subspace of finite dimension.

To address the FEM methodology, a case study presented [14] will be discussed, in which the boiler is mainly divided into the gas side and water/steam side. For the steam side, the optimal value of NF was determined to be 100 and of Nj to be 20, considering the relationship between error and computational cost. The distribution of the partitioning system can be seen in Figure 3.

For each small system, a mass balance is performed, corresponding to inputs, outputs, consumption, and generation. Under the assumption that the surfaces behave as gray bodies, the Dittus-Boelter correlation is used for the convective part. Correlations are considered to model the heat exchange between the flue gases in the hearth and the water in the casing. To simulate the water-steam system (casing), three main flows must be considered: m_f is the water feed flow, m_s is the steam outflow and m_p is the blowdown outflow. These assumptions generate several equations, which are used in conjunction with the main energy balance shown in Equation 29.

$$\frac{d}{dt} T = \frac{1}{\rho V_{total} C_v} \left[\dot{m}_f C_{pf} (T_f - T) - \dot{m}_v \lambda + \dot{q}_{F,water} + \sum_{j=2}^{3,4} \dot{q}_{j,water} \right] \quad (29)$$

Consequently, the main efficiency of the boiler is calculated according to Equation 30.

$$\eta_{boiler, l} = \frac{\sum_{j=1}^{j=Ms} (m_{vj}^+ h_{vj}^+ - m_{fj} h_{fj})}{\sum_{j=1}^{j=Ms} (m_{fuelj} * PCI)} \quad (30)$$

Where j corresponds to the simulation time interval, from 1 to the end of the simulation. The length of the time intervals remains constant (dt).

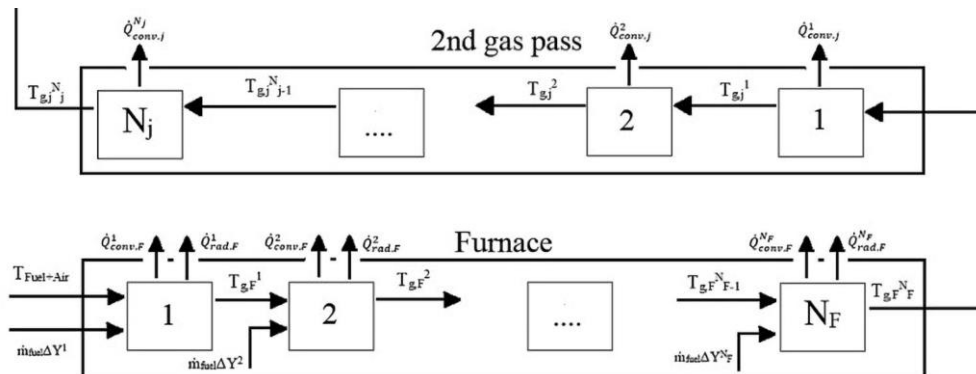


Figure 3. Structure of a FEM system for a boiler [14]

2.3.2.2 CFD

Computational Fluid Dynamic Modeling (CFD) is a complex methodology that models in detail the behavior of liquids and gases from their physicochemical properties, generating a three-dimensional model of the fluids, as shown in Figure 4[15]. The physicochemical properties are deeply linked to the temperature of the streams, so heat transfer, either by convection or radiation, is successfully simulated. However, it requires vast computational power to generate such complex systems, so their applicability is often limited to case studies or academia [14].

Accordingly, when the level of detail required warrants a CFD simulation and there is no need for real-time calculations, CFD can be applied for a specific subsystem. Among the subsystems typically modeled with this methodology are the heat exchanger and the convective heat transfer zone. The detailed CFD calculation is based on simulations of previously designed programs such as CFX, CFD ACE+, or CFD GEOM (to generate the spatial distribution mainly).

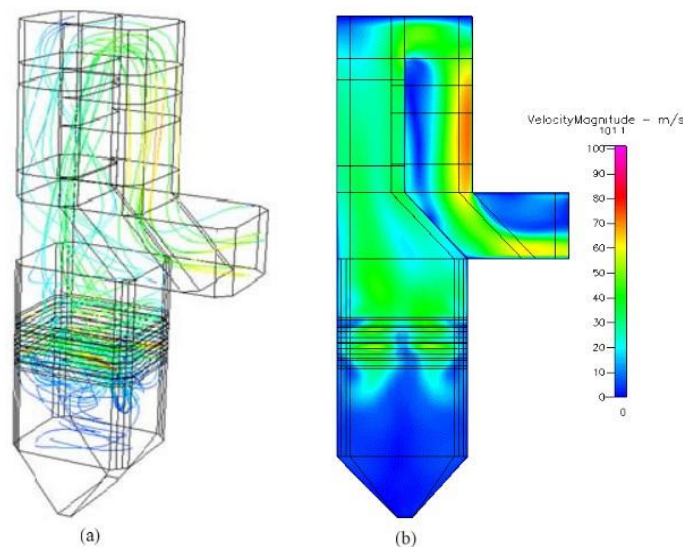


Figure 4. Example of CFD simulation. (a) Flow pattern. (b) Velocity distribution [12]

The success is to show 3D modeling's of the different systems. the geometric dimensions of what is to be simulated, the flow direction, and the different implications of the process as such are chosen. Some of the parameters such as gas composition, velocity, temperature profiles for the different flows, mass balances, energy balances, and the different assumptions are taken as an input when modeling the system, which is provided directly to the software.

2.4 Heat Transfer analysis using modeling

The review article presents a comprehensive methodology for integrated boiler process modeling, applicable to each boiler component. This framework simultaneously solves mass, energy, and momentum balance equations from inlet to outlet, utilizing component characteristic equations to define source terms. Real fluid properties referenced to the standard state are used instead of ideal gas assumptions, necessitating the use of Higher Heating Value (HHV) in energy balance equations, particularly for combustion analysis. Key component models include the adiabatic flame temperature model, which assumes instantaneous and complete combustion, except for solid fuels where partial combustion of unburned carbon is considered. This model employs real fluid properties for gas mixture analysis. The furnace model employs two zero-dimensional methods—the projected and direct methods—for furnace analysis, quantifying radiation heat transfer to water walls and direct radiation leaving the furnace exit plane, impacting downstream heat exchangers. Comparative studies show slight discrepancies in heat load predictions, with the projected method generally providing more accurate results. The radiative-convective heat exchanger model integrates convective and

radiative heat transfer mechanisms affecting heat exchanger surfaces, considering factors such as direct radiation from flue gas inside the heat exchanger cavity, impinging radiation from upstream heat exchangers, and heat absorption by exchanger walls. This model's adaptability across different geometries, such as superheaters, reheaters, economizers, and air heaters, underscores its robustness. The article emphasizes the importance of accurately calculating the effective emissivity and absorptivity of flue gas and particle suspensions, crucial for models involving high ash fuels like coal-fired boilers. Different approaches (standard, low particle load, and high particle load models) are discussed, highlighting the preference for the high particle load model due to improved accuracy, even in scenarios with lower ash loading. The methodology integrates detailed analysis with practical applicability, validated through case studies against site measurements and computational fluid dynamics (CFD) models, demonstrating its utility across diverse boiler configurations and operational conditions.[16]

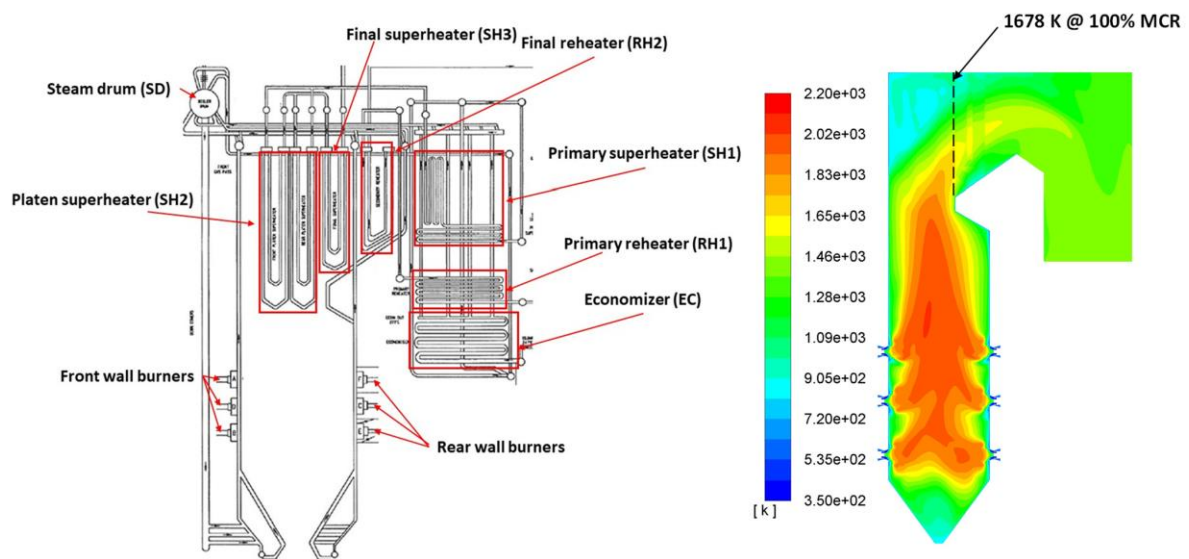


Figure 5. Utility-scale coal-fired boiler general arrangement (left) [17]; contour of gas temperatures for 100% case, calculated using CFD (right).

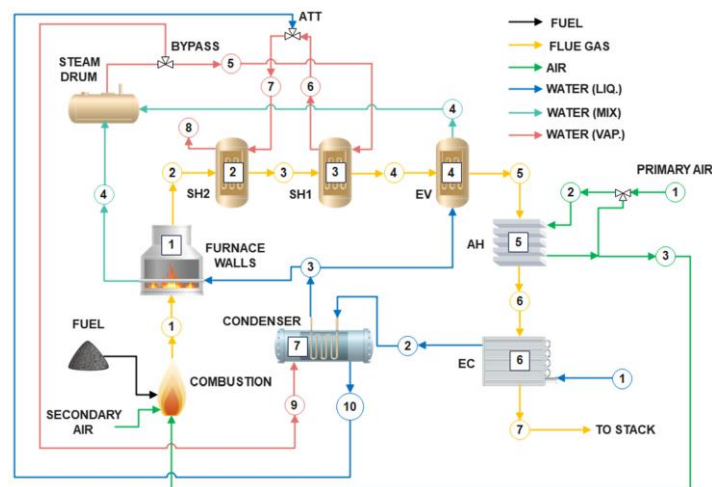


Figure 11. Process flow diagram of industrial-scale biomass-fired boiler model. [16]

3.Results and Discussion

The validation of the integrated boiler process models (M1 and M2) against site measurements taken over a two-hour period for a biomass-fired boiler operating at 100% Maximum Continuous Rating (MCR). The measured values for oxygen (O₂) and carbon dioxide (CO₂) fractions were 4.2% (O₂) and 16% (CO₂), with minimum and maximum ranges provided. The M1 model, adjusted for excess air ratio, predicted 4.2% for O₂ and 16.6% for CO₂, closely matching the measurements. Both models showed good accuracy for steam temperatures, within 0.4% at the economizer (EC) outlet and within 3.3% at the superheater 1 (SH1) outlet. M1 slightly overpredicted the flue gas temperature at the evaporator (EV) exit by 0.6%, while M2 underpredicted it by 1.0%.

For the air heater (AH) outlet, M1 overpredicted by 5.7% and M2 by 4.3%. The EC outlet temperature was underpredicted by both models, M1 by 4.8% and M2 by 5.3%. M1's prediction of the AH outlet temperature was within 0.04% of the measured value, while M2's was within 1.3%. The differences between M1 and M2 were smaller for the biomass boiler compared to the coal-fired boiler. Heat transfer rates for each heat exchanger component were also analyzed. M1 and M2 produced results mostly within the measured range, with deviations for M1 ranging from 0.9% (EC) to 4.2% (SH1), and for M2 from 1% (EC) to 10% (SH1).

M1 overpredicted the EV heat load by 1.7%, while M2 overpredicted it by 4.0%. Furnace exit gas temperatures calculated by M1 and M2 were 1206 K and 1151 K, respectively, compared to 1264 K from CFD results. M1 underpredicted by 4.6%, and M2 by 8.9%. The high particle load model proved effective even with low particle loading, supporting its use over the standard model. Heat transfer mechanisms varied by component, with direct radiation significant for SH2, while convection dominated from the EV onwards, a contrast to the coal-fired boiler where gas and particle radiation remained influential downstream.

Additional simulations at 62% MCR confirmed the models' capability to capture low-load performance, with fuel flow rate adjusted to 8.2 kg/s and attemperator flow rate to 1.75 kg/s. Heat transfer predictions at part load showed minimal error, confirming the robustness of the models. Comparing the direct and projected methods for furnace analysis, the projected method provided better overall accuracy, slightly overpredicting the furnace and EV bank heat load by 1.7%, compared to a 3% underprediction by the direct method. The findings underscore the preference for the high particle load model and the projected method for enhanced accuracy in heat transfer predictions for biomass-fired boilers. Fluid flow patterns obtained from CFD simulations provide insights into the fluid dynamics within the boiler. Identifying regions of recirculation and flow separation can help in optimizing the design to improve fluid flow and enhance combustion efficiency. For example, CFD simulations by Echi et al.[18] identified regions of recirculation and suggested design modifications to improve fluid flow

Thermal analysis of industrial boilers involves a range of methodologies, including exergy analysis, thermal efficiency optimization, CFD simulations, and detailed circulation analysis. The integration of these approaches provides a comprehensive understanding of boiler performance, enabling significant improvements in energy efficiency and cost-effectiveness. This section synthesizes the key findings from these studies. A detailed case study on heat transfer in circulating fluidized bed boilers (CFBB) by Andersson and Leckner provides valuable insights into the experimental methods [14]. The heat transfer coefficients measured in circulating fluidized bed boilers (CFBB) vary depending on the method used. For example, Andersson and Leckner [19] reported heat transfer coefficients around 130 W/m²K with a scatter of $\pm 15\%$ due to different measurement methods. These variations highlight the importance of selecting appropriate measurement techniques.

CFD has been used extensively for the analysis of solid fuel combustion systems. It can adequately resolve the complex fluid flow and heat transfer phenomena, as well as the combustion processes[20].CFD simulations provide detailed insights into the heat transfer processes within the

boiler. Rossiello et al. [21] used CFD to simulate the thermal behaviors of oil/gas boilers, finding that the heat transfer coefficients varied significantly across different regions of the boiler. Energy and Exergy Efficiencies. Energy analysis provides a measure of the boiler's efficiency based on the first law of thermodynamics. Saidur et al.[22] reported that the energy efficiencies of industrial boilers typically range from 70% to 90%, depending on the operating conditions and design Exergy analysis provides a more comprehensive evaluation of the boiler's performance by considering the quality of energy. Zhang et al.[23] conducted an exergy analysis of coal-fired industrial boilers, finding that the exergy efficiencies were significantly lower than the energy efficiencies due to irreversibility in the combustion and heat transfer processes. Chao et al.[24] investigated the impact of design modifications on the thermal efficiency of boilers. They found that optimizing the geometry and materials of the boiler walls could significantly improve the heat transfer efficiency. Regular maintenance and retrofitting with modern technologies are essential for maintaining high efficiency.

A study was conducted on a 12 MW CFBB at Chalmers University of Technology. The combustion chamber had a cross-section of 1.7 m by 1.7 m and was 13.5 m tall. The boiler walls were equipped with thermocouples and heat flow meters to measure heat transfer coefficients at various points. Heat The total heat power absorbed by the boiler walls was calculated, and the average heat transfer coefficient was determined. Local Heat Flow Meters: Heat flow meters were used to measure local heat flux directly. Thermocouples were used to measure temperatures at various points on the boiler walls. The temperature increase of the cooling water was measured to calculate the heat transfer coefficient. The average heat transfer coefficient was found to be around 130 W/m²K with a scatter of $\pm 15\%$ due to different measurement methods. The heat transfer coefficient varied with height, decreasing from 131 W/m²K at 2 m height to 93 W/m²K at 12.5 m. The variations in heat transfer coefficients highlight the importance of selecting appropriate measurement techniques. The study demonstrated that higher concentrations of particles near the wall could decrease the heat transfer coefficient due to shielding effects.

Conclusion

Thermal analysis of industrial boilers involves a range of methodologies, including exergy analysis, thermal efficiency optimization, CFD simulations, and detailed circulation analysis. The integration of these approaches provides a comprehensive understanding of boiler performance, enabling significant improvements in energy efficiency and cost-effectiveness. Ongoing research and development in this field are vital for advancing industrial boiler technology and minimizing their environmental impact.

Future Recommendations

Ongoing research and development in this field are vital for advancing industrial boiler technology and minimizing their environmental impact. The results from various studies on the thermal analysis of industrial boilers provide valuable insights into their performance and efficiency. Recent advances and innovations in boiler technology focus on improving efficiency, reducing emissions, and enhancing operational flexibility. These advancements include smart boiler technologies, advanced materials, and hybrid boiler systems.

6.1 Smart Boiler Technologies

Smart boiler technologies leverage advanced sensors, data analytics, and automation to optimize boiler performance, improve efficiency, and reduce emissions.

1. Internet of Things (IoT) Integration: IoT sensors and devices collect real-time data on boiler performance, fuel consumption, and emissions. This data is analyzed to identify trends, detect anomalies, and optimize operations

2. Predictive Maintenance: Predictive maintenance uses data analytics and machine learning algorithms to predict equipment failures and schedule maintenance activities. This approach minimizes downtime and extends the lifespan of boiler components
3. Advanced Control Systems: Smart control systems use algorithms and feedback loops to optimize combustion, adjust air-fuel ratios, and maintain optimal operating conditions. These systems enhance efficiency and reduce emissions

6.2 Advanced Materials

The development of advanced materials has led to significant improvements in the thermal performance and durability of industrial boilers. These materials offer enhanced heat resistance, corrosion resistance, and mechanical strength, making them suitable for high-temperature and high-pressure applications.

1. Superalloys: Superalloys are high-performance materials with excellent heat and corrosion resistance. They are commonly used in critical components such as boiler tubes and heat exchangers.
2. Ceramic Composites: Ceramic composites combine the properties of ceramics and metals, offering high thermal stability and mechanical strength. They are used in applications requiring resistance to extreme temperatures and harsh environments.
3. Nanomaterials: Nanomaterials, such as carbon nanotubes and graphene, have unique thermal and mechanical properties. They are being explored for use in thermal coatings and insulation materials to enhance heat transfer and reduce energy losses.

6.3 Hybrid Boiler Systems

Hybrid boiler systems combine multiple fuel sources, such as natural gas and biomass, to provide flexible and sustainable energy solutions. They offer the ability to switch between fuels based on availability and cost.

1. Design and Operation: Hybrid boilers are designed to operate with multiple fuel sources, allowing for seamless transition between fuels. This flexibility helps in optimizing fuel use and reducing operational costs.
2. Advantages and Limitations: Hybrid boilers offer significant advantages in terms of fuel flexibility and sustainability. However, their design and operation are more complex, requiring advanced control systems to manage the transition between fuels.

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